

COGNITIVE RADIO SIMULATION - VIDEO DEMO SCRIPT

Factual, balanced talking track for recording a demonstration. Covers what the software does today, without over-claiming.

- Software: eAge Cognitive Radio Channel Selection under Jamming (Streamlit PC demo)
- Login (if shown): Username pocuser | Password poc123
- Suggested length: about 8-12 minutes

Before you start (30 seconds)

- Confirm the app is running and you are logged in.
- Use Dark theme if that matches the rest of your recording.
- Prefer System Architecture & Node View first; switch to Operator Spectrum View only if you want a denser map.
- Start with no jam so the map shows a calm FREE baseline, then inject jam.

Stage 1 - Opening and scope (about 1 minute)

Say:

1. This is a software-only Cognitive Radio simulation for technical review.
2. It does not radiate RF; it is a PC demonstration of sensing, decision, and actuation logic.
3. It is prepared as a demo for DRDO Labs and related defence technical audiences.
4. The goal is to show how a link can observe channel quality, decide (rules / ML / hybrid), and act (channel change, hop set, or power advice) under jamming profiles.

Do not claim:

- Live EW deployment readiness
- Real I/Q capture from field radios in this build
- Formal certified train/test ML metrics beyond what the UI shows

Stage 2 - Screen layout (about 1-1.5 minutes)

Point to the header:

1. Title: channel selection under jamming.
2. Short capability line: channel simulation, measurement-based receiver judgement, Cognitive Radio decision, adaptive hopping, power advice.
3. eAge branding on the right.

Sidebar - keep it practical:

1. User - signed-in session; Logout ends the demo session.
2. Display View - Architecture/Node view vs Operator Spectrum view.
3. Theme - Light / Dark (top-right on the main canvas).
4. Configuration - number of channels (N), decision engine, hopping toggle, SINR thresholds, max TX power.
5. Run control - single step, auto steps, reset.

Architecture view (if visible):

1. Separate nodes for jammer, transmitter, receiver / sensing, and Cognitive Radio decision.
2. Useful when explaining the closed loop as blocks, not only as a colour map.

Stage 3 - Benign baseline (about 1 minute)

Do:

1. Ensure jam is clear / profile NONE.
2. Run Single step once or twice if needed.
3. Show the channel map: FREE / DEGRADED / BLOCKED, with the active TX channel marked.

Say:

1. With no ECM applied, the simulator gives a benign baseline - active carrier in a usable (FREE) condition for the demo.
2. Receiver judgement is measurement-based on simulated SINR and jam power, not a hand-tuned storyboard.
3. States are simple and readable for review: FREE, DEGRADED, BLOCKED.

Stage 4 - Inject jamming (about 2 minutes)**Do (pick 2-3; do not rush all):**

1. Spot - jam one channel (ideally the current TX channel) -> Apply jam -> Single step.
2. Multi-spot - jam two channels -> Apply -> Step.
3. Optionally Sweep or Barrage briefly to show wider pressure.

Say:

1. Jam profiles are controlled scenarios inside the software: Spot, Multi-spot, Sweep, Barrage (and clear).
2. After apply, the map and metrics update: SINR drops where jam is present; states move toward DEGRADED / BLOCKED as appropriate.
3. This is how we exercise the Cognitive Radio loop under contested spectrum in simulation.

If Architecture view: use the Jammer node controls; mention sidebar jam controls appear in Operator view.

Stage 5 - Decision engines (about 2 minutes)**RULES**

1. Deterministic baseline: prefer free channels; fall back by policy when needed.
2. Transparent for scientists who want an explainable non-ML reference.

ML

1. Supervised MLP recommender (small network, e.g. 64x32) trained on synthetic observation vectors from this same simulator.
2. Features include per-channel SINR, state encoding, jam power, TX power, and power headroom (for N=8 this is a 26-length style vector as shown in the UI).
3. Output is a classification action: recommend a channel, or advise INCREASE_POWER.
4. Training accuracy shown in the UI is primarily on the synthetic training set; a formal held-out test split is a known next hardening item - do not oversell generalisation.

HYBRID (recommended for demo)

1. ML proposes; safety rules can override unsafe suggestions (e.g. blocked channels).
2. This is the balance we emphasise for defence review: learning proposal with containment.

Do:

1. Set Decision engine to HYBRID.
2. With jam on the active channel, Single step and point to the decision / ML proposal / dispatched action on screen.
3. If a safety override message appears, explain it calmly as intentional containment.

Stage 6 - Hopping and power (about 1-1.5 minutes)**Do:**

1. Enable Cognitive / adaptive hopping if you want to show hop-set behaviour; Apply configuration if required.
2. Step again under jam and point to hop-related UI / Stage trace if shown.
3. Under heavy pressure (e.g. Barrage), note INCREASE_POWER advice when headroom remains - still within configured max TX power.

Say:

1. Hopping here means the software builds / uses an adaptive hop set from sensed conditions - simulation logic, not a live frequency hopper on air.
2. Power advice is bounded by the configured maximum TX power in the demo.

Stage 7 - Explainability trace (about 1 minute)

Do:

- Open or scroll the staged explainability / Stage 1-5 style panel if visible in Architecture view.

Say (keep short):

1. Stage flow is meant to be reviewable: sense -> decide (rules/ML/hybrid) -> safety -> hop synthesis -> actuator command.
2. This supports technical discussion without treating the neural net as a black box for the whole system.

Stage 8 - Closing (about 1 minute)

Say:

1. Summary: configurable multi-channel simulation, measurement-based states, rules / ML / hybrid decisions, optional hopping, power advice, under Spot / Multi / Sweep / Barrage scenarios.
2. Interfaces are structured so a later path toward eADM / SDR integration can reuse the decision pattern - that is a roadmap statement, not a claim that this PC build is already connected to field hardware.
3. Happy to take questions on thresholds, jam models, ML features, or hybrid safety behaviour.

Optional logout (5-10 seconds):

1. Sidebar Logout -> confirm dialog -> return to login.
2. Only if you want to show session control; otherwise end on the live demo screen.

Suggested on-screen sequence (checklist)

Step	Action	Speak about
1	Login (if needed)	Demo access for the review session
2	Show header + views	Scope: software simulation
3	Clear jam, step	FREE baseline
4	Spot jam on TX CH	Contested channel
5	HYBRID + step	ML + safety
6	Multi-spot or Barrage	Different ECM pressure
7	Hopping / power (optional)	Actuation options
8	Explainability stages	Reviewability
9	Close	Honest limits + next steps

Phrases to prefer / avoid

Prefer

- In this simulation...
- Measurement-based within the model...
- Synthetic training data for the demonstrator...
- Hybrid mode keeps a safety gate around the ML proposal...
- Structured for later eADM connection without redesigning the decision interface...

Avoid

- Operational EW system / field-proven / real-time RF warfare
- Fully trained / validated ML for deployment
- Detects any jammer automatically in theatre
- Absolute performance guarantees (BER, range, classification % in the field)

One-line closing (optional)

This demonstration shows a closed-loop Cognitive Radio decision path under jamming in software - sensing, rules and ML with hybrid safety, and bounded actuation - as a technical baseline for further defence evaluation.

Prepared by: eAge Innovations Technical Engineering Team

For scientific evaluation by: Defence Scientists / DRDO Labs